

Computational Microelectronics

Lecture 1

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Course

- Computational Microelectronics
 - Code: EC7114 (/ SE7114)
 - Lecture 3, no experiment, credit 3
- Instructor: Sung-Min Hong
 - Department of EECS

Resources

- Presentation materials

<https://github.com/hi2ska2/cm2026f>

- Announcement
 - GIST LMS system

- YouTube channel

<https://www.youtube.com/@TCADHong>

Evaluation

- Attendance (10%)
- Homework (40%)
 - **New! Prepare and submit your own presentation.**
 - It will be uploaded in my YouTube channel.
- Final presentation (50%)
 - Prepare and submit your own presentation.
 - It will be uploaded in my YouTube channel.

Planned business trips

- Three weeks
 - Palma de Mallorca (September 7 and 9, ESSERC)
 - Kumamoto (September 28 and 30, SISPAD)
 - San Francisco (December. IEDM) → Term project
- Pre-recorded lectures will be uploaded.

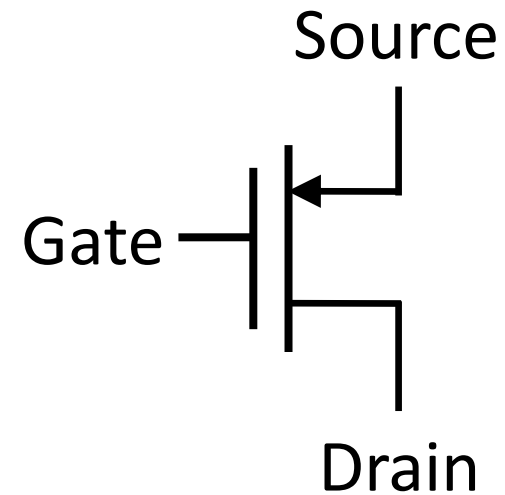
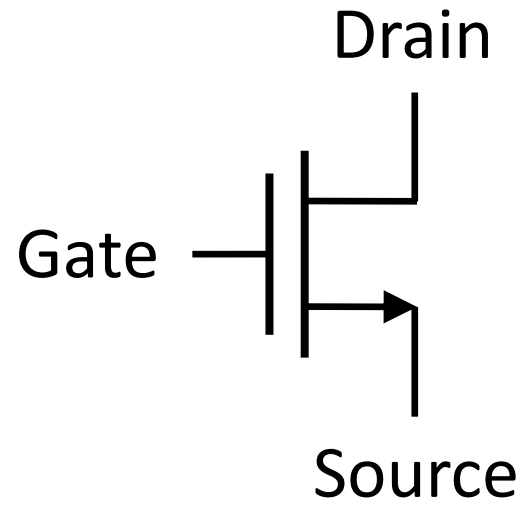
Homework#1

- Due: AM08:00, September 2
- Problem#1
 - “Hello, world!”
 - Send me your homework video by e-mail.

L1

MOSFET

- Basic unit of microelectronics
 - Controlled by the gate voltage
 - NMOSFET and PMOSFET



Level 1 model (NMOSFET)

- A simple model (Shichman-Hodges)

- Its basic current model is:

- Cutoff region, $V_{gs} < V_t$

$$I_d = 0$$

- Linear region, $V_{ds} < V_{gs} - V_t$

$$I_d = \textcolor{red}{KP} \frac{W_{eff}}{L_{eff}} (1 + \textcolor{red}{LAMBDA} \times V_{ds}) \left(V_{gs} - V_t - \frac{V_{ds}}{2} \right) V_{ds}$$

- Saturation region, $V_{ds} > V_{gs} - V_t$

$$I_d = \frac{\textcolor{red}{KP}}{2} \frac{W_{eff}}{L_{eff}} (1 + \textcolor{red}{LAMBDA} \times V_{ds}) (V_{gs} - V_t)^2$$

- (Red-colored quantities are the SPICE model parameters.)

Level 1 model (NMOSFET)

- A simple model (Shichman-Hodges)

- Threshold voltage:

- For nonnegative V_{sb} ,

$$V_t = VTO + GAMMA \left(\sqrt{PHI + V_{sb}} - \sqrt{PHI} \right)$$

- VTO is the “zero-bias” threshold voltage.

- $GAMMA$ is the body effect coefficient.

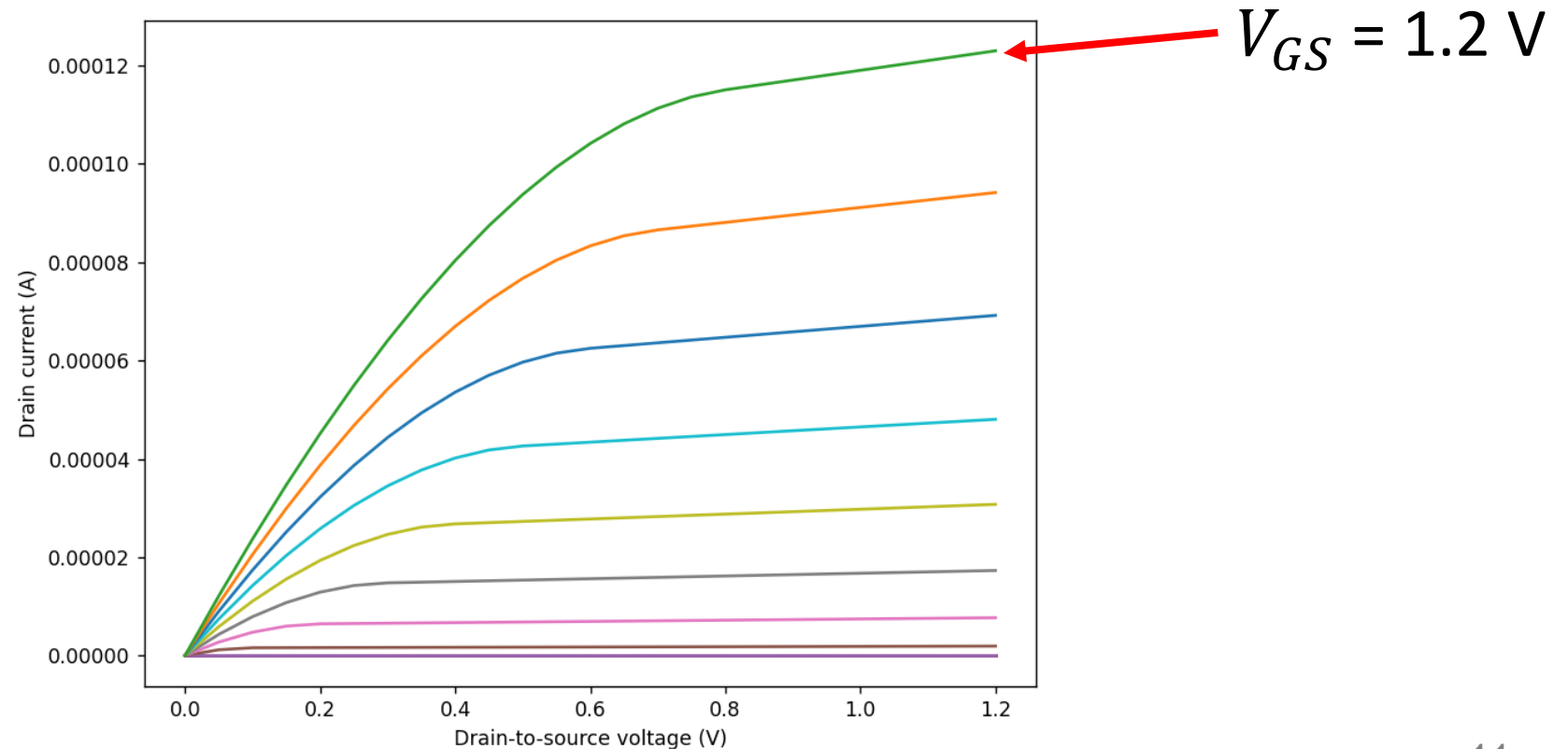
- PHI is the surface potential.

NMOSFET example

- Consider the following parameters:

$KP=155e-6$ $LAMBDA=0.2$ $VTO=0.4$ $PHI=0.93$ $GAMMA=0.6$

$$-W_{eff}/L_{eff} = 2$$

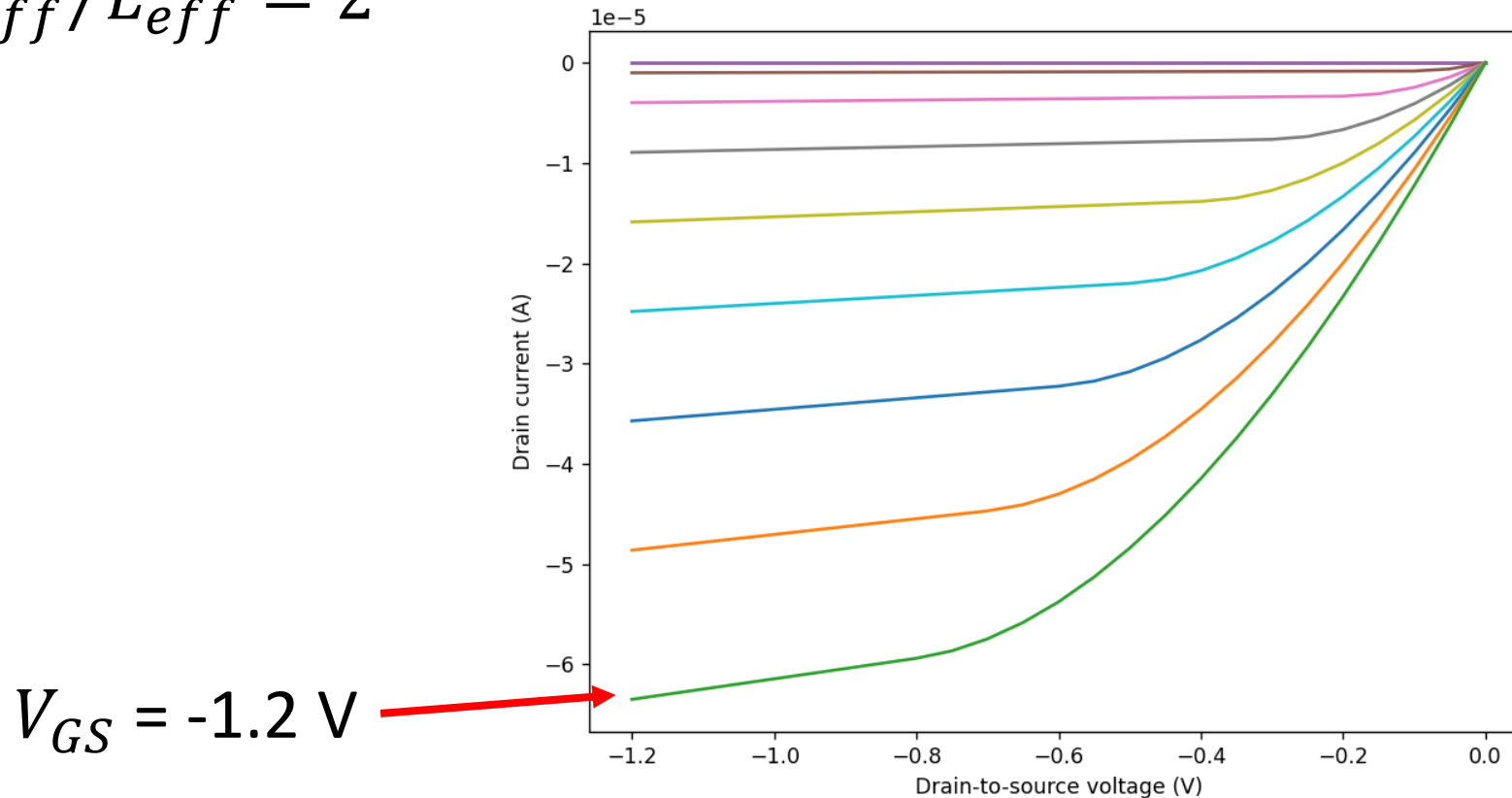


PMOSFET example

- Consider the following parameters:

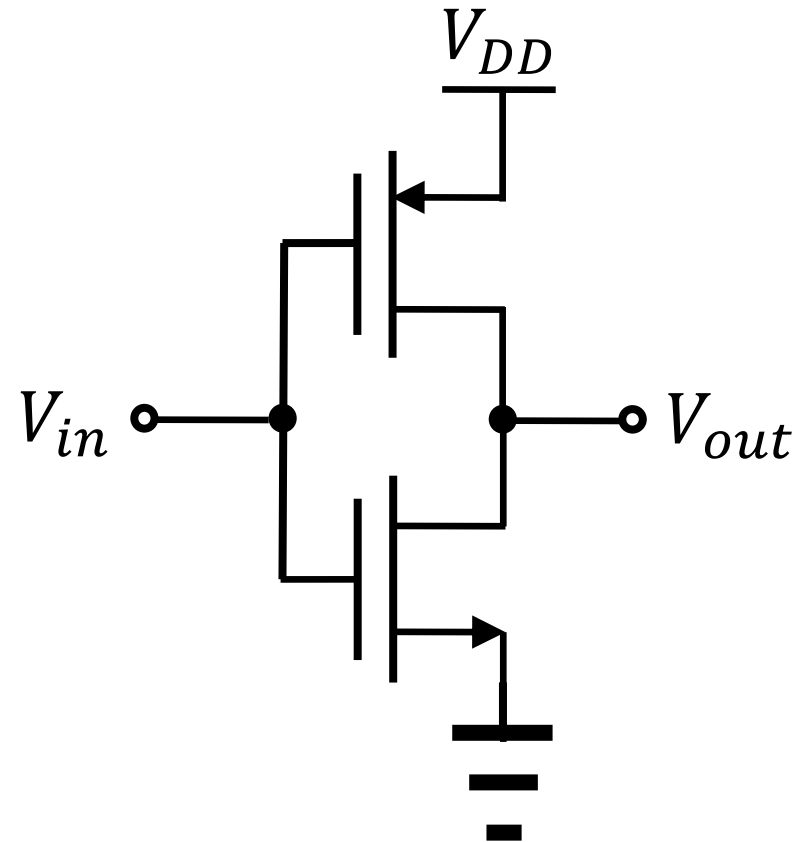
$KP=80e-6$ $LAMBDA=0.2$ $VTO=-0.4$ $PHI=0.93$ $GAMMA=0.6$

$$-W_{eff}/L_{eff} = 2$$



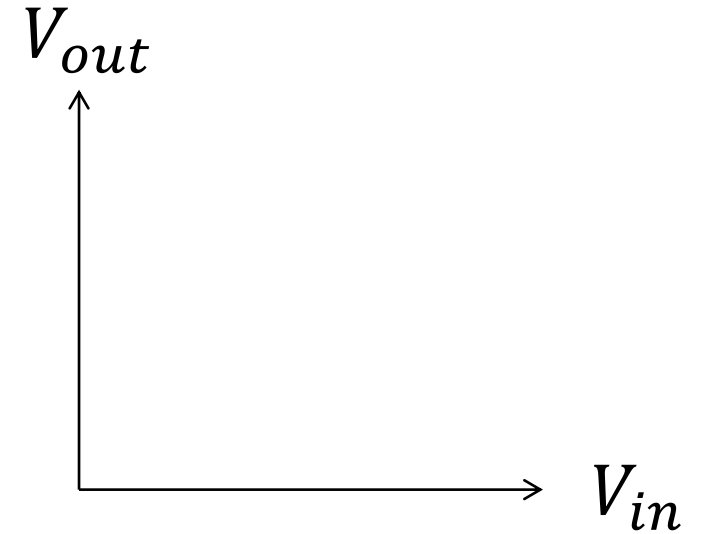
The simplest logic gate

- When $V_{in} = 0$,
 - NMOS OFF
 - PMOS ON
 - V_{out} is “pulled up.”
 - $V_{out} = V_{DD}$
- When $V_{in} = V_{DD}$,
 - NMOS ON
 - PMOS OFF
 - V_{out} is “pulled down.”
 - $V_{out} = 0$



Assume that V_{in} is given.

- The goal is to calculate V_{out} .
 - $V_{out}(V_{in})$ is called the voltage transfer curve.



- What is the condition for a point on the VTC?

$$I_{Dn} + I_{Dp} = 0$$

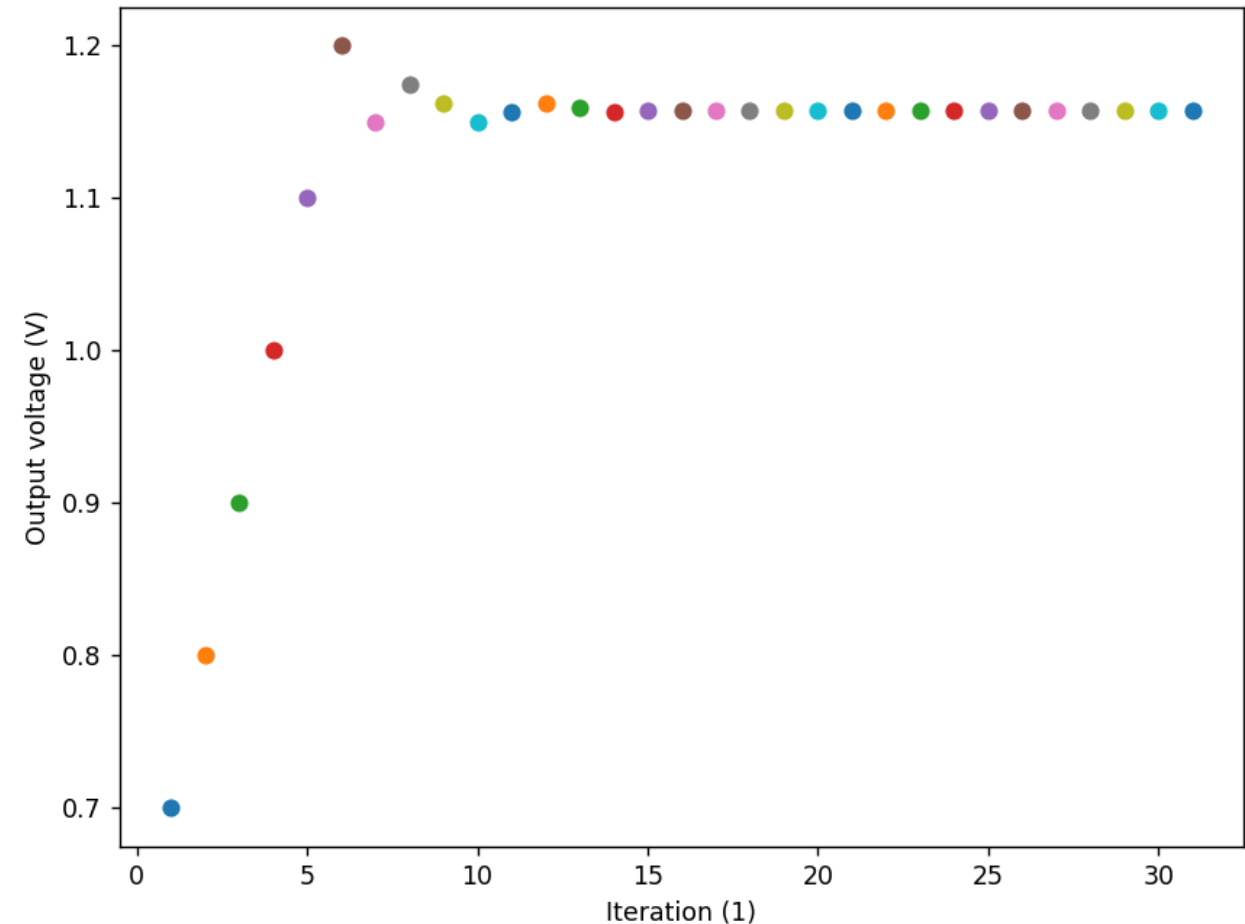
- Assume that $V_{DD} = 1.2$ V.
 - For example, at $V_{in} = 0.5$ V and $V_{out} = 0.7$ V, we have $I_{Dn} = 1.77$ μ A and $I_{Dp} = -7.92$ μ A. Certainly, it is not on the VTC.

One possible way (Trial-and-error)

- In order to increase I_{Dn} , we can try a higher V_{out} , for example 0.8 V.
 - Still, the currents are much different.
 - 0.9 V? No. 1.0 V? No. 1.1 V? No.
 - 1.2 V? Well, now, $I_{Dn} = 1.92 \mu\text{A}$ and $I_{Dp} = 0 \text{ A}$. We must increase $|I_{Dp}|$.
 - 1.15 V? $I_{Dn} = 1.91 \mu\text{A}$ and $I_{Dp} = -2.22 \mu\text{A}$. We must increase I_{Dn} .
 - 1.175 V? $I_{Dn} = 1.91 \mu\text{A}$ and $I_{Dp} = -1.16 \mu\text{A}$. We must increase $|I_{Dp}|$.
 - 1.1625 V? $I_{Dn} = 1.91 \mu\text{A}$ and $I_{Dp} = -1.70 \mu\text{A}$. We are getting closer...

Number of iterations

- When we have a voltage change smaller than 1 μV , the calculation stops.
 - Start at $\frac{V_{DD}}{2}$.
 - Initial step is 0.1 V.
 - When the sign changes, the step is reduced by a factor of 0.5.
 - Otherwise, keep the step.
 - 31 iterations at $V_{in} = 0.5 \text{ V}$



Thank you!